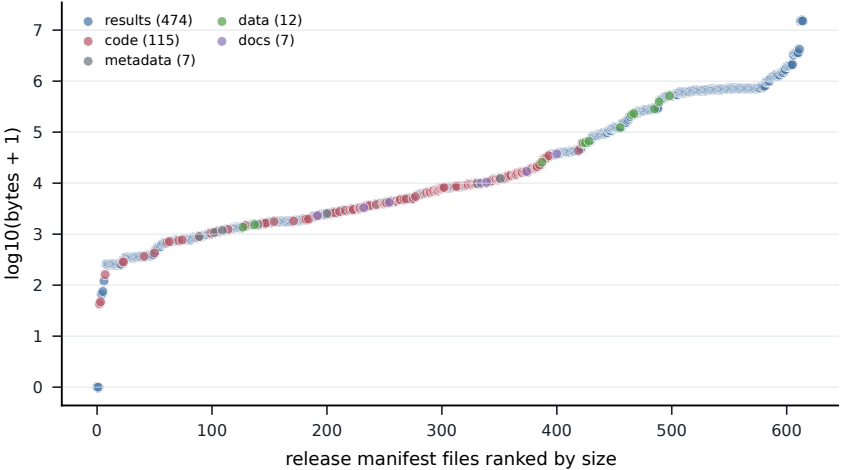


Fig. 14 | Reproducibility and release map
Source data, scripts, checksums, environment and Zenodo release form a reproducible handoff.

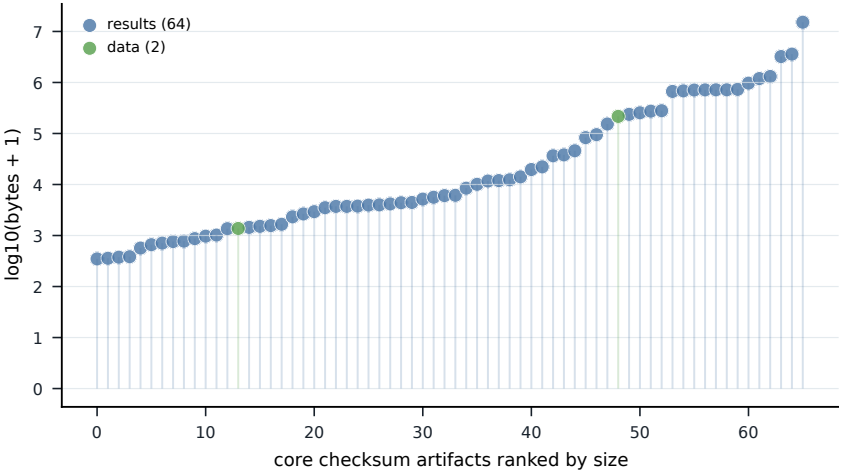
a Release manifest size spectrum



b Release verification matrix

bundle manifest exists	PASS	
bundle exists	PASS	
bundle bytes match	PASS	expected=104284966; actual=104...
bundle sha256 match	PASS	expected=dd849c78a8b90d9ae8f5d...
tar readable	PASS	members=617
upload manifest in bundle	PASS	results/nature_release/release...
zenodo metadata in bundle	PASS	results/nature_release/zenodo_...
upload manifest row count	PASS	expected=615; actual=615
upload manifest members present	PASS	missing=0
upload manifest member hashes	PASS	mismatch=0
critical artifacts present	PASS	missing=0
critical artifact hashes	PASS	mismatch=0
deposition result exists	PASS	
deposition bundle matches manifest	PASS	release_status=zenodo_publishe...
zenodo doi recorded	PASS	release_status=zenodo_publishe...
zenodo release published	PASS	release_status=zenodo_publishe...

c Checksum-covered core artifacts



d

Release identity and environment

DOI status	published
DOI	10.5281/zenodo.20550855
bundle MB	104.3
tar members	617
manifest rows	615
critical artifacts	73
torch	2.6.0
numpy	2.0.0
pandas	2.3.3
scanpy	1.12.1
anndata	0.12.16